

SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : 192424226 Name: A Roja

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Visualization Critique & Redesign

Weightage: 20%

Code of Conduct:

I A Roja [192424226] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A Roja

Criteria	Evaluation of Existing Visualization 25%	Redesign Strategy 25%	Application of Visualization Principles 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							

Q2							
Q3							
Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

Question 1 – Critique and Redesign an ECDF and Q-Q Plot Dataset:

Salary_Distribution.csv

Dataset:

Employee_ID	Department	Salary
E01	IT	42000
E02	IT	45000
E03	IT	48000
E04	IT	51000
E05	IT	55000
E06	HR	38000
E07	HR	40000
E08	HR	42000
E09	HR	46000

E10	HR	50000
E11	Finance	47000
E12	Finance	49000
E13	Finance	52000
E14	Finance	58000
E15	Finance	65000
E16	Marketing	36000
E17	Marketing	39000
E18	Marketing	43000
E19	Marketing	47000
E20	Marketing	72000

Assessment Task

A salary analyst has created an ECDF and a Q-Q plot to study employee salary distribution, but the charts use unclear labels, inappropriate scaling, and excessive decoration.

1. Create the two original-style plots in R.
2. Identify at least three visualization problems.
3. Redesign the plots using appropriate scales, labels, reference lines, and minimal visual clutter.
4. Explain how the redesigned plots improve interpretation of skewness, outliers, and normality.
5. State whether a Q-Q plot alone is sufficient to conclude that the data are normally distributed.

PSEUDOCODE:

- BEGIN
- Load salary dataset
- Store Employee ID, Department and Salary
- Calculate salary in thousands
- FOR original ECDF

Plot Salary on X-axis

Calculate cumulative proportion

Display ECDF

END FOR

- FOR original Q-Q plot
- Compare observed salary values with theoretical normal values

Plot Q-Q points

Add reference line

END FOR

- Identify visualization problems
- FOR redesigned ECDF

Use clear salary scale

Label X-axis as Salary

Label Y-axis as Cumulative Proportion

Remove unnecessary decoration

Display ECDF

END FOR

- FOR redesigned Q-Q plot

Use theoretical normal quantiles

Plot observed salary quantiles

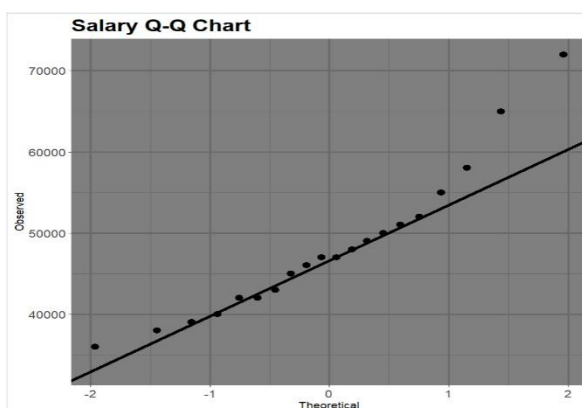
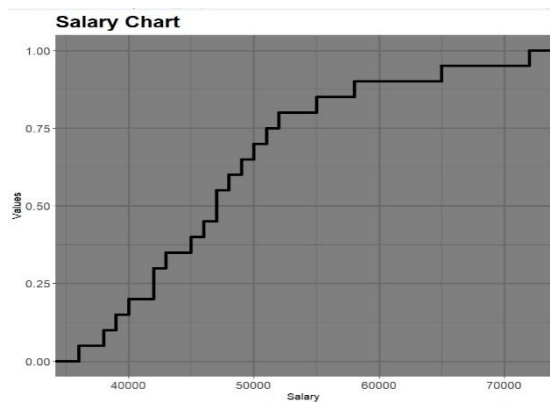
Add reference line

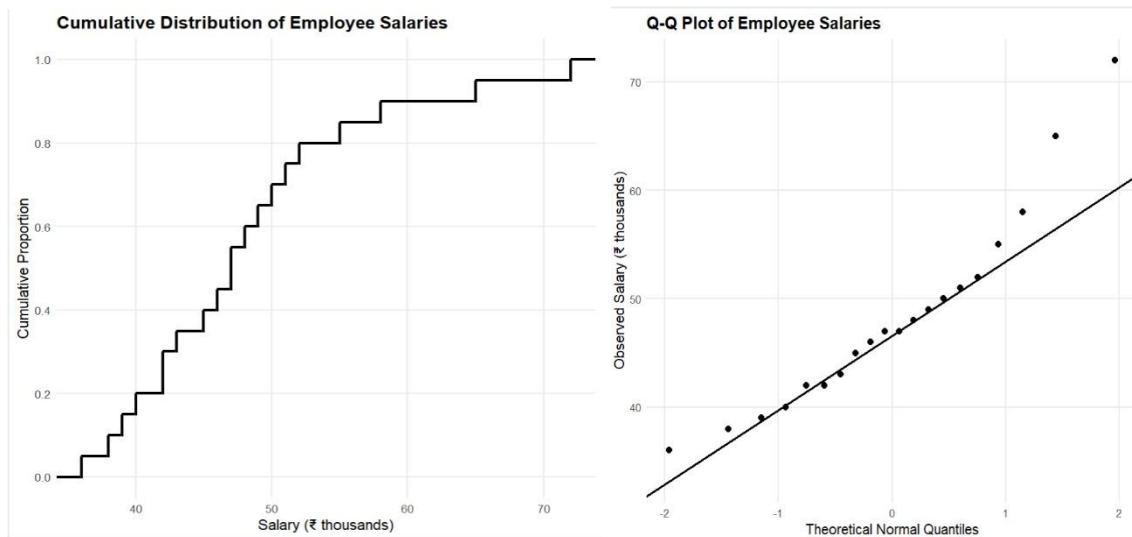
Use clear labels

END FOR

- Perform Shapiro-Wilk normality test
- Interpret skewness, outliers and normality
- END

OUTPUT:





Question 2 – Critique and Redesign Many Distributions Using Bean Plots

Dataset: Student_Scores.csv

Dataset:

Student_ID	Course	Score
S01	Python	62
S02	Python	68
S03	Python	74
S04	Python	81
S05	Python	88
S06	Statistics	58
S07	Statistics	64
S08	Statistics	71
S09	Statistics	79
S10	Statistics	86
S11	DataViz	55
S12	DataViz	61
S13	DataViz	69
S14	DataViz	77

S15	DataViz	84
S16	AI	65
S17	AI	72
S18	AI	78
S19	AI	87
S20	AI	93

Assessment Task

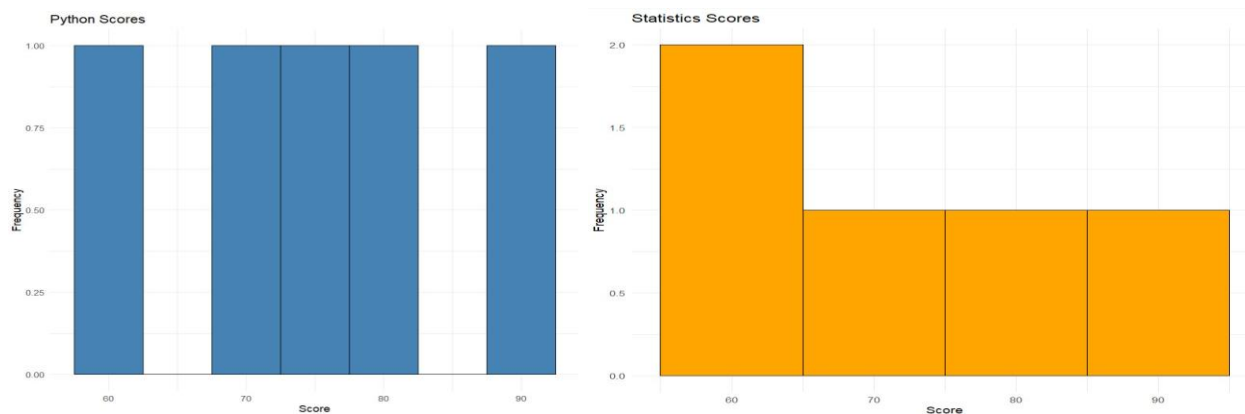
A college report uses four separate histograms with different bin widths to compare course scores. The comparison is difficult because the scales and layouts are inconsistent.

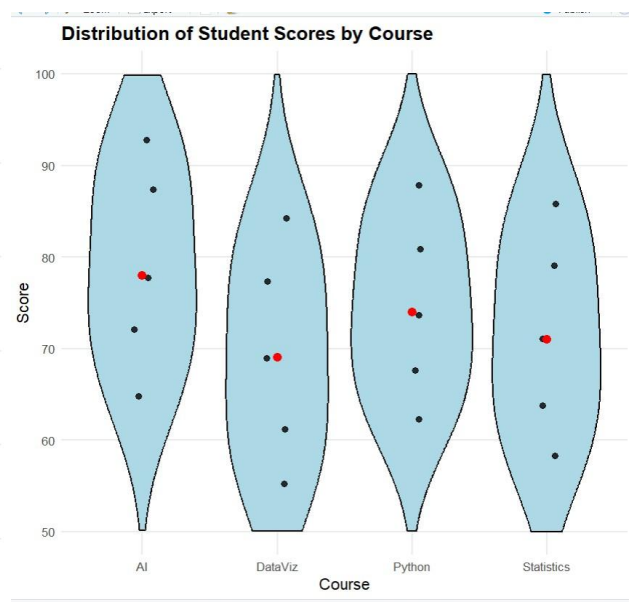
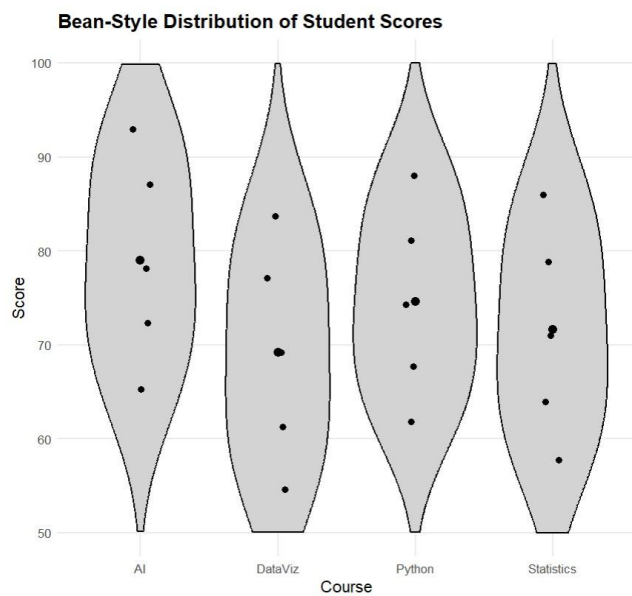
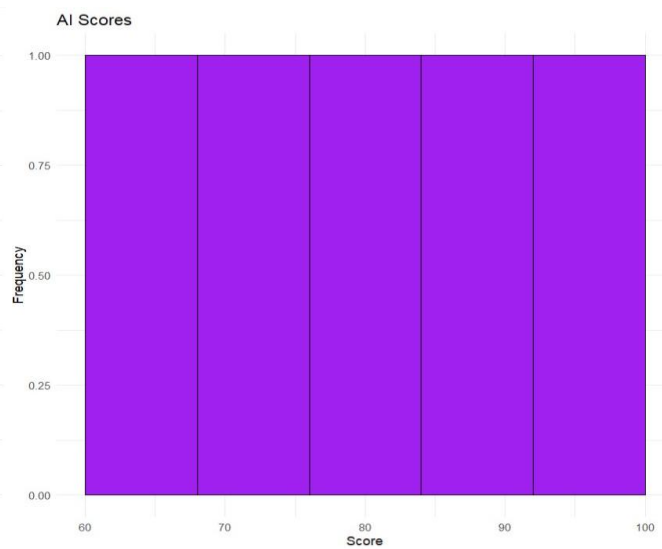
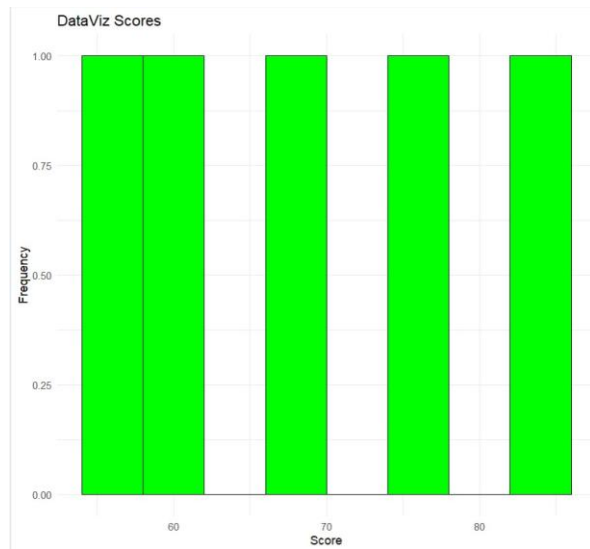
1. Reproduce the problematic visualization using R.
2. Critique the use of separate histograms, inconsistent scales, and bin widths.
3. Redesign the visualization using bean plots or a suitable violin/density-based alternative.
4. Add an appropriate common scale and clear course labels.
5. Explain which course has the highest concentration of scores and which course shows the greatest spread.

PSEUDOCODE:

- START
- Load Student Scores dataset
- Group scores according to courses
- Create original histograms
- Identify problems with bin widths and scales
- Create bean or violin plot
- Use a common score scale and clear labels
- Compare concentration and spread of courses
- END

OUTPUT:





Question 3 – Critique and Redesign Proportion Visualizations

Dataset: Regional_Purchases.csv Dataset:

Region	Product	Purchases
North	Laptop	25
North	Phone	35
North	Tablet	20
South	Laptop	30
South	Phone	40

South	Tablet	25
East	Laptop	28
East	Phone	32
East	Tablet	18
West	Laptop	22
West	Phone	38
West	Tablet	27

Assessment Task

A manager presents product proportions using four pie charts, one for each region. The charts use different starting angles, 3-D effects, and inconsistent labeling.

1. Create a representative version of the misleading visualization.
2. Identify at least four design problems.
3. Redesign the analysis using side-by-side bars and a 100% stacked bar chart.
4. Compare the two redesigns and determine which is better for comparing regional proportions. 5. Explain why 3-D effects and inconsistent scales can lead to misleading conclusions.

PSEUDOCODE:

- START
- Load Regional Purchases dataset
- Calculate product proportions for each region
- Create original pie charts
- Identify problems such as 3-D effects and inconsistent angles
- Create side-by-side bar chart
- Create 100% stacked bar chart
- Compare regional product proportions
- Select the better visualization
- END

OUTPUT:



Question 4 – Critique and Redesign Nested Proportions

Dataset: Sales_Hierarchy.csv

Dataset:

Region	Category	Subcategory	Sales
North	Electronics	Laptop	52000
North	Electronics	Phone	38000
North	Furniture	Chair	22000
North	Furniture	Desk	18000
South	Electronics	Laptop	46000
South	Electronics	Phone	42000
South	Furniture	Chair	25000

South	Furniture	Desk	21000
East	Electronics	Laptop	58000
East	Electronics	Phone	35000
East	Furniture	Chair	19000
East	Furniture	Desk	23000
West	Electronics	Laptop	49000
West	Electronics	Phone	44000
West	Furniture	Chair	27000
West	Furniture	Desk	20000

Assessment Task

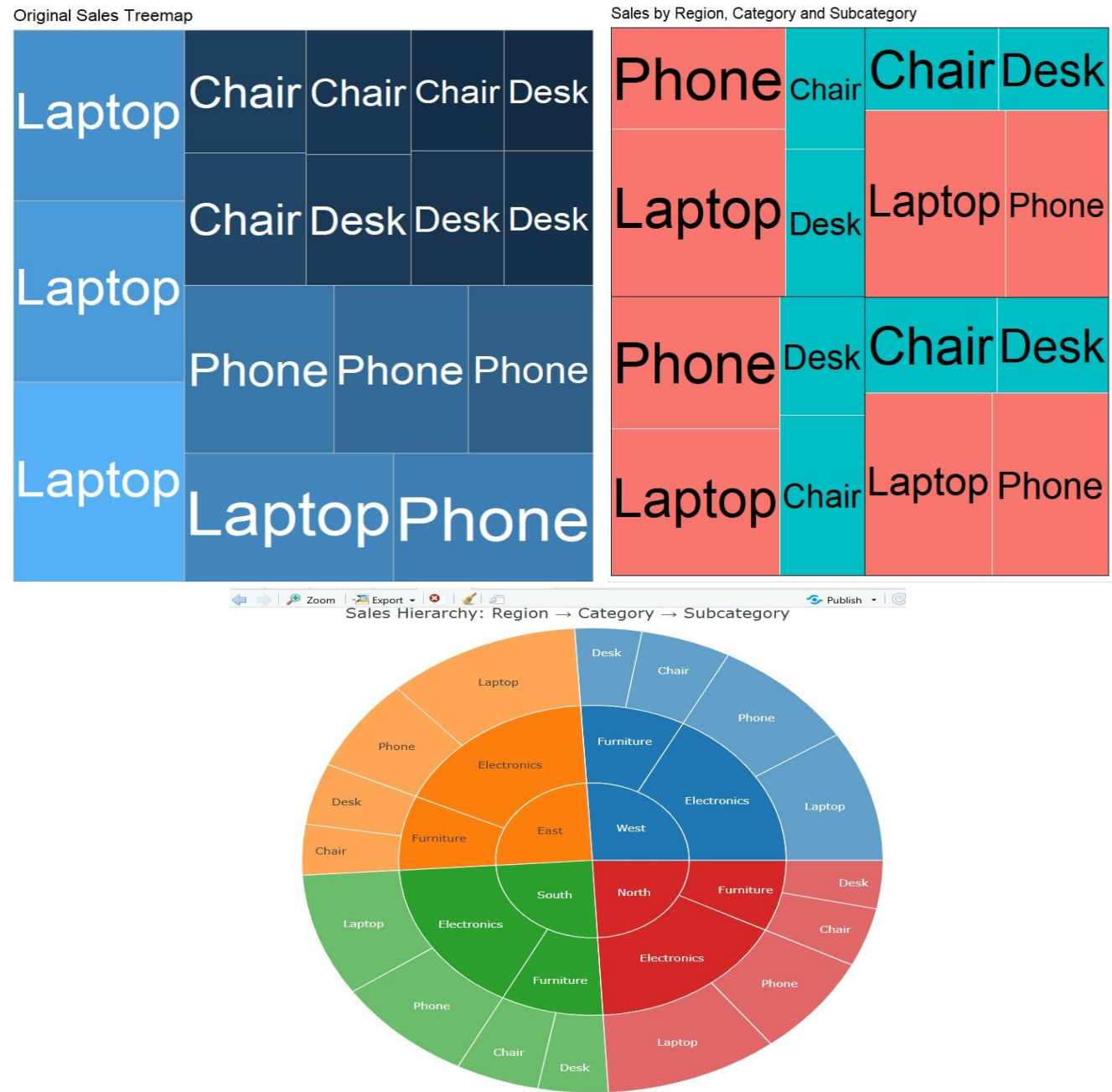
An organization uses a treemap to communicate Region → Category → Subcategory sales. Small areas have unreadable labels and the color scale does not clearly communicate the hierarchy.

1. Create the original treemap.
2. Critique the size, labeling, color, and hierarchy encodings.
3. Redesign the visualization as an improved treemap.
4. Create a sunburst alternative and compare it with the redesigned treemap.
5. Recommend one visualization for a management presentation and justify your choice.

PSEUDOCODE:

- START
- Load Sales Hierarchy dataset
- Create original treemap
- Identify problems in size, color, labels and hierarchy
- Create redesigned treemap
- Create Region → Category → Subcategory sunburst
- Compare treemap and sunburst
- Find highest-selling region and category
- Recommend the suitable visualization
- END

OUTPUT:



Question 5 – Critique and Redesign Flow-Based Proportions

Dataset: Career_Flow.csv

Dataset:

Specialization	Internship	Placement_Sector	Students
Data Analytics	Yes	IT	18

Data Analytics	Yes	Finance	12
Data Analytics	No	IT	8
AI	Yes	IT	20
AI	Yes	Healthcare	10
AI	No	IT	6
Cyber Security	Yes	IT	16
Cyber Security	Yes	Finance	9
Cyber Security	No	Government	7
Cloud Computing	Yes	IT	15
Cloud Computing	Yes	Finance	8
Cloud Computing	No	Government	5

Assessment Task

A placement office uses a Sankey diagram to show Specialization → Internship → Placement Sector. The original chart uses excessively similar colors, unlabelled flows, and proportions that are difficult to compare. 1. Create a Sankey diagram representing the data. 2. Identify at least four potential sources of confusion or misleading interpretation.

3. Redesign the Sankey diagram with meaningful labels, consistent flow representation, and an appropriate visual hierarchy.
4. Create a parallel sets alternative and compare the two designs.
5. Explain which visualization better communicates the major student flows and why.

PSEUDOCODE:

- START
- Load Career Flow dataset
- Create Specialization → Internship → Placement flow
- Create original Sankey diagram
- Identify confusing colors, labels and flow proportions
- Redesign Sankey with clear labels and spacing
- Create Parallel Sets/Alluvial plot
- Compare major student flows
- Recommend the better visualization
- END

OUTPUT

